

Group-A & B

(a) Discuss complex number and complex conjugate.

- **Complex Number:** A complex number is a number of the form $z = x + iy$, where x and y are real numbers, and i is the imaginary unit defined by $i^2 = -1$. Here, x is called the **real part** of z , denoted as $\text{Re}(z)$, and y is called the **imaginary part** of z , denoted as $\text{Im}(z)$.
 - Example: For $z = 3 + 2i$, $\text{Re}(z) = 3$ and $\text{Im}(z) = 2$.
 - Example: For $z = 2 - (1/2)i$, $\text{Re}(z) = 2$ and $\text{Im}(z) = -1/2$.
- **Complex Conjugate:** The complex conjugate of a complex number $z = x + iy$ is denoted by $\bar{z}$ (or sometimes z^*) and is defined as $\bar{z} = x - iy$. It is obtained by changing the sign of the imaginary part.
 - Example: If $z = 3 + 2i$, then $\bar{z} = 3 - 2i$.
 - Example: If $z = 2 - 3i$, then $\bar{z} = 2 + 3i$. ↓
 - Example: If $z = 4i$, then $\bar{z} = -4i$.
 - Example: If $z = 5$ (a real number), then $\bar{z} = 5$.

(b) Derive the relationship between the cartesian and polar form of a complex number. Express

$(1+2i)/(1-3i)$ and $(2+3i)/(3-7i)$ in the polar form.

• Relationship:

A complex number $z = x + iy$ can be represented geometrically as a point (x, y) in the Cartesian plane (Argand diagram). The distance from the origin to this point is the modulus $r = |z| = \sqrt{x^2 + y^2}$. The angle θ that the line segment from the origin to (x, y) makes with the positive real axis is the argument $\arg(z)$.

Using trigonometry:

$$x = r \cos \theta$$

↓

$$y = r \sin \theta$$

Therefore, $z = x + iy = r \cos \theta + i r \sin \theta = r(\cos \theta + i \sin \theta)$. This is the **polar form** of the complex number. It can also be written using Euler's formula as $z = re^{i\theta}$.

• Expressing $(1+2i)/(1-3i)$ in Polar Form:

1. Simplify the expression:

Multiply numerator and denominator by the conjugate of the denominator $(1+3i)$:

$$\frac{(1+2i)(1+3i)}{(1-3i)(1+3i)} = \frac{[1 + 3i + 2i + 6i^2]}{[1 - (3i)^2]} = \frac{[1 + 5i - 6]}{[1 - (-9)]} = \frac{[-5 + 5i]}{10} = -1/2 + (1/2)i$$

2. Find Modulus (r) and Argument (θ):

Let $w = -1/2 + (1/2)i$.

$$\blacksquare \text{ Modulus: } r = |w| = \sqrt{(-1/2)^2 + (1/2)^2} = \sqrt{1/4 + 1/4} = \sqrt{1/2} = 1/\sqrt{2}.$$

- **Argument:** Since w is in the second quadrant (negative real, positive imaginary),

$$\theta = \pi - \tan^{-1}(|\operatorname{Im}(w)/\operatorname{Re}(w)|) = \pi - \tan^{-1}((1/2)/(1/2)) = \pi - \tan^{-1}(1) = \pi - \pi/4 = 3\pi/4.$$

3. Write in Polar Form:

$$w = (1/\sqrt{2})(\cos(3\pi/4) + i \sin(3\pi/4)) \text{ or } w = (1/\sqrt{2})e^{i3\pi/4}.$$

- **Expressing $(2+3i)/(3-7i)$ in Polar Form:**

1. Simplify the expression:

Multiply numerator and denominator by the conjugate of the denominator $(3+7i)$:

$$\frac{[(2+3i)(3+7i)]}{[(3-7i)(3+7i)]} = \frac{[6 + 14i + 9i + 21i^2]}{[9 - (7i)^2]} = \frac{[6 + 23i - 21]}{[9 - (-49)]} = \frac{[-15 + 23i]}{58} = -15/58 + (23/58)i$$

2. Find Modulus (r) and Argument (θ):

$$\text{Let } w = -15/58 + (23/58)i.$$

- **Modulus:** $r = |w| = \sqrt{(-15/58)^2 + (23/58)^2} = (1/58)\sqrt{225 + 529} = (1/58)\sqrt{754}.$

- **Argument:** Since w is in the second quadrant,

$$\theta = \pi - \tan^{-1}(|\operatorname{Im}(w)/\operatorname{Re}(w)|) = \pi - \tan^{-1}((23/58)/(15/58)) = \pi - \tan^{-1}(23/15).$$

3. Write in Polar Form:

$$w = (\sqrt{754}/58)(\cos(\pi - \tan^{-1}(23/15)) + i \sin(\pi - \tan^{-1}(23/15))) \text{ or}$$

$$w = (\sqrt{754}/58)e^{i(\pi - \tan^{-1}(23/15))}.$$

(c) Define absolute value and argument of a complex number.

- **Absolute Value (Modulus):** The absolute value (or modulus) of a complex number $z = x + iy$ is denoted by $|z|$ and is defined as $|z| = \sqrt{x^2 + y^2}$. Geometrically, it represents the distance from the origin to the point (x, y) representing z in the Argand diagram.
 - Example: $|3 + 4i| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$
 - Example: $|3 - 4i| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$
- **Argument:** The argument of a non-zero complex number $z = x + iy$ is denoted by $\arg(z)$ or θ and is defined as the angle θ (in radians) that the line segment joining the origin to the point (x, y) makes with the positive real axis. It is given by $\theta = \tan^{-1}(y/x)$, but the specific quadrant must be considered to determine the correct angle. The principal value of the argument usually lies in the interval $(-\pi, \pi]$ or $[0, 2\pi)$.
 - Example: For $z = 2 + 2\sqrt{3}i$, $\arg(z) = \tan^{-1}((2\sqrt{3})/2) = \tan^{-1}(\sqrt{3}) = \pi/3$ (since it's in the first quadrant).

(d) For any two complex numbers z_1 and z_2 , prove that the argument of the product of two complex numbers is equal to the sum of their argument.

- **Proof:**

Let $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$ be the polar forms of the complex numbers.

Their product is:

$$z_1 z_2 = (r_1 e^{i\theta_1})(r_2 e^{i\theta_2}) = r_1 r_2 e^{i(\theta_1 + \theta_2)} = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

The argument of $z_1 z_2$ is the angle in its polar form, which is $\theta_1 + \theta_2$.

Therefore, $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$.

(e) For any two complex numbers z_1 and z_2 , prove that the modulus of the product of two complex numbers is equal to the product of their moduli.

- **Proof:**

Let $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$ be the polar forms of the complex numbers.

Their product is:

$$z_1 z_2 = (r_1 e^{i\theta_1})(r_2 e^{i\theta_2}) = r_1 r_2 e^{i(\theta_1 + \theta_2)}$$

The modulus of $z_1 z_2$ is the magnitude $r_1 r_2$.

The modulus of z_1 is r_1 and the modulus of z_2 is r_2 .

Therefore, $|z_1 z_2| = r_1 r_2 = |z_1| |z_2|$.

(f) Show that the ellipse $|z + 3| + |z - 3| = 10$ can be expressed in rectangular form as $x^2/25 + y^2/16 = 1$.

- **Proof:**

Let $z = x + iy$.

The equation becomes: $|(x + 3) + iy| + |(x - 3) + iy| = 10$.

This means: $\sqrt{(x + 3)^2 + y^2} + \sqrt{(x - 3)^2 + y^2} = 10$.

Let $A = \sqrt{(x + 3)^2 + y^2}$ and $B = \sqrt{(x - 3)^2 + y^2}$, so $A + B = 10$.

Rearrange: $A = 10 - B$.

Square both sides: $A^2 = (10 - B)^2 = 100 - 20B + B^2$.

Substitute back A^2 and B^2 :

$$(x + 3)^2 + y^2 = 100 - 20B + (x - 3)^2 + y^2$$

Expand: $x^2 + 6x + 9 + y^2 = 100 - 20B + x^2 - 6x + 9 + y^2$.

Simplify: $6x = 100 - 20B - 6x$.

Rearrange: $12x - 100 = -20B$.



Divide by 4: $3x - 25 = -5B$.

So, $B = (25 - 3x)/5$.

Now square both sides again: $B^2 = [(25 - 3x)/5]^2$.

Substitute $B^2 = (x - 3)^2 + y^2$:

$$(x - 3)^2 + y^2 = (25 - 3x)^2 / 25.$$

Expand both sides:

Left: $x^2 - 6x + 9 + y^2$.

Right: $(625 - 150x + 9x^2) / 25 = 25 - 6x + (9/25)x^2$.

So: $x^2 - 6x + 9 + y^2 = 25 - 6x + (9/25)x^2$.

Subtract $-6x$ from both sides: $x^2 + 9 + y^2 = 25 + (9/25)x^2$.

Rearrange: $x^2 - (9/25)x^2 + y^2 = 25 - 9$.

Combine terms: $(16/25)x^2 + y^2 = 16$.



Divide both sides by 16: $(16/25)x^2/16 + y^2/16 = 1$.

Simplify: $x^2/25 + y^2/16 = 1$.

Hence proved.

(g) Prove that $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2|z_1|^2 + 2|z_2|^2$.

• **Proof:**

Recall that for any complex number z , $|z|^2 = z * \bar{z}$ (where $\bar{z}$ is the conjugate).

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$.

$$\text{LHS} = |z_1 + z_2|^2 + |z_1 - z_2|^2$$

$$= (z_1 + z_2)(\bar{z}_1 + \bar{z}_2) + (z_1 - z_2)(\bar{z}_1 - \bar{z}_2)$$

$$= [z_1\bar{z}_1 + z_1\bar{z}_2 + z_2\bar{z}_1 + z_2\bar{z}_2] + [z_1\bar{z}_1 - z_1\bar{z}_2 - z_2\bar{z}_1 + z_2\bar{z}_2]$$

$$\begin{aligned}
&= z_1 \bar{z}_1 + z_1 \bar{z}_2 + z_2 \bar{z}_1 + z_2 \bar{z}_2 + z_1 \bar{z}_1 - z_1 \bar{z}_2 - z_2 \bar{z}_1 + z_2 \bar{z}_2 \\
&= 2z_1 \bar{z}_1 + 2z_2 \bar{z}_2 \\
&= 2|z_1|^2 + 2|z_2|^2 \\
&= \text{RHS.}
\end{aligned}$$

Hence proved.

(h) Find the locus represented by (i) $|z - 3 + i| = 4$ (ii) $|z| = 0$.

- (i) $|z - 3 + i| = 4$:

Let $z = x + iy$. Then $|(x - 3) + i(y + 1)| = 4$.

This means $\sqrt{(x - 3)^2 + (y + 1)^2} = 4$.

Squaring both sides: $(x - 3)^2 + (y + 1)^2 = 16$.

This is the equation of a circle with center at $(3, -1)$ and radius 4 .

- (ii) $|z| = 0$:

Let $z = x + iy$. Then $\sqrt{x^2 + y^2} = 0$.

This implies $x^2 + y^2 = 0$, which is only possible if $x = 0$ and $y = 0$.

Therefore, the locus is the single point at the origin $(0, 0)$.

(i) The center of a regular hexagon is at the origin and one vertex is given by $\sqrt{3} + i$ on the Argand diagram. Find the complex number represented by the other vertices.

- Solution:

A regular hexagon has 6 vertices, equally spaced around a circle centered at the origin. The angle between consecutive vertices is $2\pi/6 = \pi/3$ radians.

Given vertex: $z_0 = \sqrt{3} + i$.



First, find its modulus and argument to understand its position.

$$|z_0| = \sqrt{(\sqrt{3})^2 + 1^2} = \sqrt{3 + 1} = \sqrt{4} = 2.$$

$$\arg(z_0) = \tan^{-1}(1/\sqrt{3}) = \pi/6 \text{ (since it's in the first quadrant).}$$

So, $z_0 = 2(\cos(\pi/6) + i \sin(\pi/6))$.

The other vertices are found by rotating z_0 by multiples of $\pi/3$:

◦ Vertex 1 (given): $z_0 = 2(\cos(\pi/6) + i \sin(\pi/6)) = \sqrt{3} + i$

◦ Vertex 2: $z_1 = 2(\cos(\pi/6 + \pi/3) + i \sin(\pi/6 + \pi/3)) = 2(\cos(\pi/2) + i \sin(\pi/2)) = 2(0 + i*1) = 2i$

◦ Vertex 3:

$$z_2 = 2(\cos(\pi/6 + 2\pi/3) + i \sin(\pi/6 + 2\pi/3)) = 2(\cos(5\pi/6) + i \sin(5\pi/6)) = 2(-\sqrt{3}/2 + i*1/2) = -\sqrt{3} + i$$

o Vertex 4:

$$z_3 = 2(\cos(\pi/6 + \pi) + i \sin(\pi/6 + \pi)) = 2(\cos(7\pi/6) + i \sin(7\pi/6)) = 2(-\sqrt{3}/2 - i*1/2) = -\sqrt{3} - i$$

o Vertex 5:

$$z_4 = 2(\cos(\pi/6 + 4\pi/3) + i \sin(\pi/6 + 4\pi/3)) = 2(\cos(3\pi/2) + i \sin(3\pi/2)) = 2(0 - i*1) = -2i$$

o Vertex 6:

$$z_5 = 2(\cos(\pi/6 + 5\pi/3) + i \sin(\pi/6 + 5\pi/3)) = 2(\cos(11\pi/6) + i \sin(11\pi/6)) = 2(\sqrt{3}/2 - i*1/2) = \sqrt{3} - i$$

The other five vertices are: $2i$, $-\sqrt{3} + i$, $-\sqrt{3} - i$, $-2i$, $\sqrt{3} - i$.

(j) Show that the general equation of a real circle in an Argand plane is of the form

$$z \bar{z} + a z + \bar{a} \bar{z} + b = 0.$$

• Proof:

Let $z = x + iy$ and $\bar{z} = x - iy$.

The general equation of a circle in the Cartesian plane is $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center and r is the radius.

Expanding: $x^2 - 2hx + h^2 + y^2 - 2ky + k^2 = r^2$.

Rearrange: $x^2 + y^2 - 2hx - 2ky + (h^2 + k^2 - r^2) = 0$.

Note that $x^2 + y^2 = |z|^2 = z \bar{z}$.

Also, $x = (z + \bar{z})/2$ and $y = (z - \bar{z})/(2i) = -i(z - \bar{z})/2$.

Substitute:

$$z \bar{z} - 2h*(z + \bar{z})/2 - 2k*(-i(z - \bar{z})/2) + (h^2 + k^2 - r^2) = 0.$$

Simplify:

$$z \bar{z} - h(z + \bar{z}) + ik(z - \bar{z}) + (h^2 + k^2 - r^2) = 0.$$

$$z \bar{z} - hz - h\bar{z} + ikz - ik\bar{z} + (h^2 + k^2 - r^2) = 0.$$

Group terms:

$$z \bar{z} + (-h + ik)z + (-h - ik)\bar{z} + (h^2 + k^2 - r^2) = 0.$$

Let $a = -h + ik$ and $b = h^2 + k^2 - r^2$. Note that $\bar{a} = -h - ik$.

So, the equation becomes: $z \bar{z} + a z + \bar{a} \bar{z} + b = 0$.

Hence proved.

(k) Explain analytic function. Show that the function $f(z) = e^x(\cos y + i \sin y)$ is analytic, also find its derivative.

- **Analytic Function:** A function $f(z)$ is said to be analytic at a point z_0 if it is differentiable not only at z_0 but also in some neighborhood of z_0 . A function is analytic in a domain D if it is analytic at every point in D . For a function $f(z) = u(x, y) + iv(x, y)$ to be analytic, the Cauchy-Riemann equations must be satisfied: $\partial u / \partial x = \partial v / \partial y$ and $\partial u / \partial y = -\partial v / \partial x$, and the partial derivatives must be continuous.

- **Show $f(z) = e^x(\cos y + i \sin y)$ is analytic:**

Here, $z = x + iy$, so $f(z) = e^x \cos y + i e^x \sin y$.

Thus, $u(x, y) = e^x \cos y$ and $v(x, y) = e^x \sin y$.

Compute partial derivatives:

- $\partial u / \partial x = e^x \cos y$
- $\partial u / \partial y = -e^x \sin y$
- $\partial v / \partial x = e^x \sin y$
- $\partial v / \partial y = e^x \cos y$



Check Cauchy-Riemann:

- $\partial u / \partial x = e^x \cos y = \partial v / \partial y$ ✓
- $\partial u / \partial y = -e^x \sin y = -(\partial v / \partial x) = -(e^x \sin y)$ ✓

The partial derivatives are continuous everywhere (as exponential and trig functions are smooth).

Therefore, $f(z)$ satisfies the Cauchy-Riemann equations and has continuous partial derivatives, so it is analytic everywhere in the complex plane.

- **Find its derivative:**

For an analytic function, $f'(z) = \partial u / \partial x + i \partial v / \partial x$.

$$f'(z) = e^x \cos y + i (e^x \sin y) = e^x (\cos y + i \sin y).$$

Notice that $f'(z) = f(z)$. (This is expected because $f(z) = e^x e^{iy} = e^{(x+iy)} = e^z$, and the derivative of e^z is e^z).

(I) Check the function $f(z) = z^3 + 2 \operatorname{Re}\{z^2 + z \bar{z} + 1\} + i \operatorname{Im}\{|z|^2\}$ is analytic or not, also find its derivative.

- **Check Analyticity:**

Let $z = x + iy$, so $\bar{z} = x - iy$.

First, simplify the expression:

- $z^2 = (x + iy)^2 = x^2 - y^2 + 2ixy$
- $z \bar{z} = |z|^2 = x^2 + y^2$
- $z^2 + z \bar{z} + 1 = (x^2 - y^2 + 2ixy) + (x^2 + y^2) + 1 = 2x^2 + 1 + 2ixy$
- $\operatorname{Re}\{z^2 + z \bar{z} + 1\} = 2x^2 + 1$
- $|z|^2 = x^2 + y^2$
- $\operatorname{Im}\{|z|^2\} = \operatorname{Im}\{x^2 + y^2\} = 0$ (since $x^2 + y^2$ is real)

- $\text{Im}\{|z|^2\} = \text{Im}\{x^2 + y^2\} = 0$ (since $x^2 + y^2$ is real)

So, $f(z) = z^3 + 2(2x^2 + 1) + 0 = z^3 + 4x^2 + 2$.

Now, write $f(z)$ in terms of x and y :

$$z^3 = (x + iy)^3 = x^3 + 3x^2(iy) + 3x(iy)^2 + (iy)^3 = x^3 + 3ix^2y - 3xy^2 - iy^3 = (x^3 - 3xy^2) + i(3x^2y - y^3)$$

Therefore, $f(z) = (x^3 - 3xy^2) + i(3x^2y - y^3) + 4x^2 + 2 = (x^3 - 3xy^2 + 4x^2 + 2) + i(3x^2y - y^3)$.

So, $u(x, y) = x^3 - 3xy^2 + 4x^2 + 2$ and $v(x, y) = 3x^2y - y^3$.

Compute partial derivatives:

- $\partial u / \partial x = 3x^2 - 3y^2 + 8x$
- $\partial u / \partial y = -6xy$
- $\partial v / \partial x = 6xy$
- $\partial v / \partial y = 3x^2 - 3y^2$



Check Cauchy-Riemann:

- $\partial u / \partial x = 3x^2 - 3y^2 + 8x$
- $\partial v / \partial y = 3x^2 - 3y^2$

These are **not equal** unless $8x = 0$, i.e., $x = 0$.

- $\partial u / \partial y = -6xy$
- $-\partial v / \partial x = -6xy$

These are equal.

Since $\partial u / \partial x \neq \partial v / \partial y$ (except on the line $x=0$), the Cauchy-Riemann equations are not satisfied in any open set. Therefore, $f(z)$ is **not analytic** anywhere.

- **Derivative:**

Since the function is not analytic, it does not have a derivative in the complex sense (except possibly at isolated points where C-R might hold, but not in a neighborhood). We cannot define a complex derivative $f'(z)$ for this function.

Group-C

(a) Explain harmonic function

- **Harmonic Function:** A real-valued function $\phi(x, y)$ of two real variables x and y is called a **harmonic function** if it has continuous second-order partial derivatives and satisfies **Laplace's equation**:

$$\nabla^2 \phi = \partial^2 \phi / \partial x^2 + \partial^2 \phi / \partial y^2 = 0.$$



Harmonic functions are important in complex analysis because if $f(z) = u(x, y) + iv(x, y)$ is analytic, then both $u(x, y)$ and $v(x, y)$ are harmonic functions. They are also fundamental in physics, describing phenomena like gravitational potential, electrostatic potential, and steady-state temperature distribution.

(b) See the problem of harmonic function properly.

This instruction seems to mean you should study problems related to harmonic functions thoroughly.

Common problems include:

- Verifying if a given function is harmonic.
- Finding a harmonic conjugate $v(x, y)$ for a given harmonic function $u(x, y)$ such that $f(z) = u + iv$ is analytic.
- Verifying if a given function is harmonic.
- Finding a harmonic conjugate $v(x, y)$ for a given harmonic function $u(x, y)$ such that $f(z) = u + iv$ is analytic.
- Solving Laplace's equation with boundary conditions (Dirichlet or Neumann problems).
- Using properties of harmonic functions (e.g., mean value property, maximum principle).

Make sure you understand how to check harmonicity and how to find harmonic conjugates using the Cauchy-Riemann equations.